Java Minecraft Server First Plugin

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Time Required: 60 minutes

Minecraft Server Plugins

One of the great things about Minecraft is its open extensibility. Anyone with a knowledge of Java can create mods and plugins.

We are going to create a Spigot heal plugin.

1: Minecraft Server Setup

To test our Java plugins, we need a local Minecraft server that supports plugins. We will use Paper, https://papermc.io/ , a high speed popular fork of Bukkit.

1. If you don't have the latest Amazon Corretto JDK 21 → https://docs.aws.amazon.com/corretto/latest/corretto-21-ug/downloads-list.html
2. Create a folder for your Minecraft server.
3. Go to www.serverjars.com → go to **Auto Updater** → Download **ServerJars.jar**
4. Place this file in your server folder.
5. Create the following batch file named **StartServer.bat** to start your server.

```
java -Xms2G -Xmx2G -jar ServerJars.jar
```

6. Double Click the batch file. A configuration screen will start up.
7. Would you like to create a config file now? **Y**
8. Which server category would you like to use? **servers**
9. Which server type would you like to use? **purpur**

10. What server version would you like to run? **Latest**

11. Would you like to use always the same server jar for every ServerJars instance? **N**

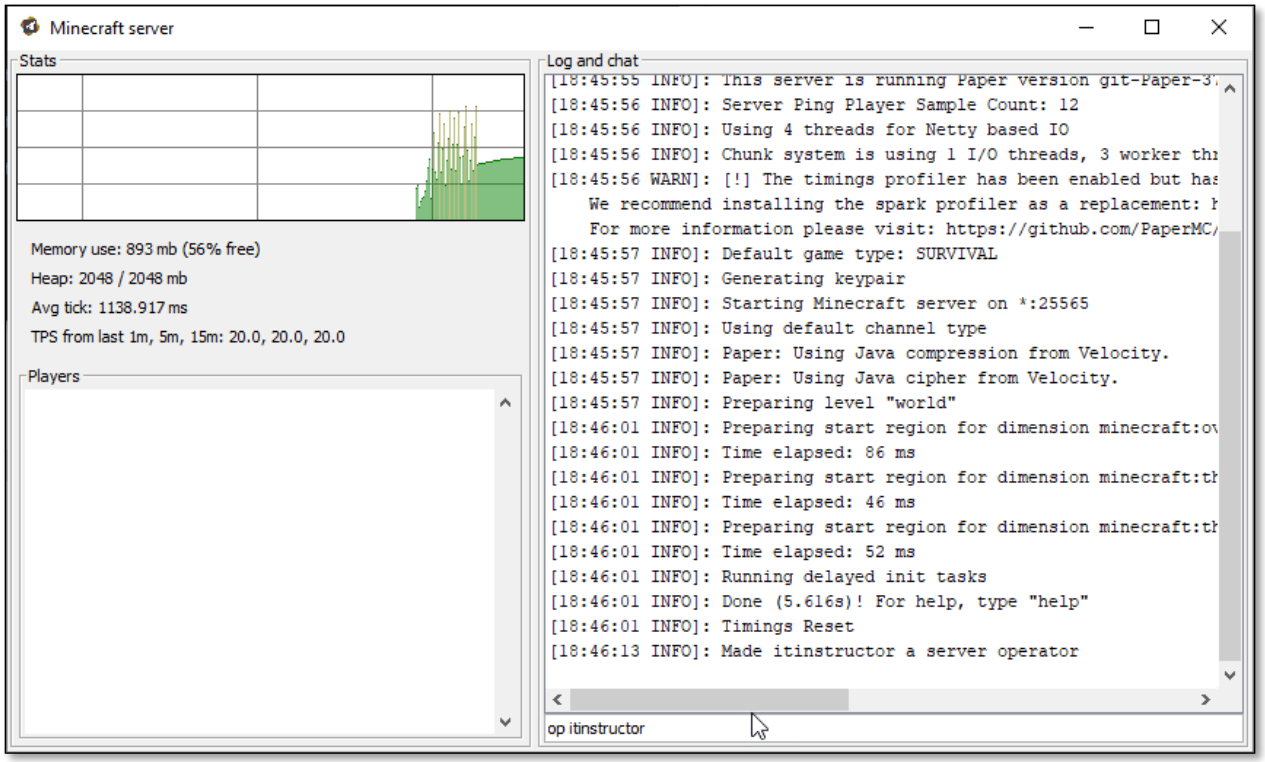
12. The server will download and install.

13. You will see a message about agreeing to the EULA. The server will shutdown.

14. Open the text file **eula.txt**

15. Change **eula=true**

16. Start the server. It should load this time.



17. Add yourself as op in the server GUI as show above: **op username**

18. Close the GUI to stop the server.

2: Configure the Server

1. Open **server.properties**

    a. Change **gamemode=creative**

2. Start the server.

3. Open your Minecraft client. **Multiplayer → Add Server**
    a. Server Name: **Server Tutorial**
    b. Server Address: **localhost:25565**
    c. Click Done.
4. Click the Play button.
5. Open chat: **/gamemode creative**
6. You are in creative mode and free to explore your new server world!

3: Install IntelliJ IDEA

IntelliJ is a Java IDE that has a lot of nice built in features to make it easy to build plugins. There is an open source Community edition.

As a student, you can get access to the full version of IntelliJ and any of their other development tools, such as PyCharm.
https://www.jetbrains.com/community/education/#students

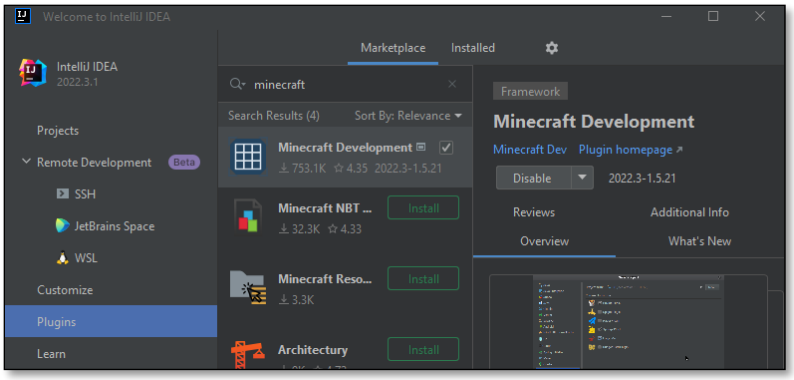
An open source built Minecraft Development IntelliJ plugin makes creating Minecraft plugins and mods pretty straightforward.

1. Go to https://www.jetbrains.com/idea/
2. Click **Downloads** → download and install the **Community Edition**.

4: Create Heal Plugin

Our first plugin will be a heal plugin.

1. Start IntelliJ. You will see Welcome to IntelliJ IDEA.
2. Click **Plugins** → Search for **Minecraft** → Click **Install**.



1. Go to **Projects** → **New Projects** → Choose **Minecraft** → **Spigot Plugin**.
2. **Project SDK**: You should see Project SDK: Amazon Coretto version 21. Click **Next**.
3. Build Settings.
    a. GroupId: **edu.wncc**
    b. ArtifactId: **heal**
4. Click **Next**. **Spigot Settings** → leave as is. Click **Next**.
5. Project name: **HealTutorial**
6. Browse to a good location to save the project.
7. Click **Create**. Your project will take a few moments to build. It will automatically open your **Heal.java** file.
8. Go to **src** → **main** → **resources**. Double Click **plugin.yml** to edit the file.
9. Add the code below to the end of this file.

```
commands:
  heal:
```

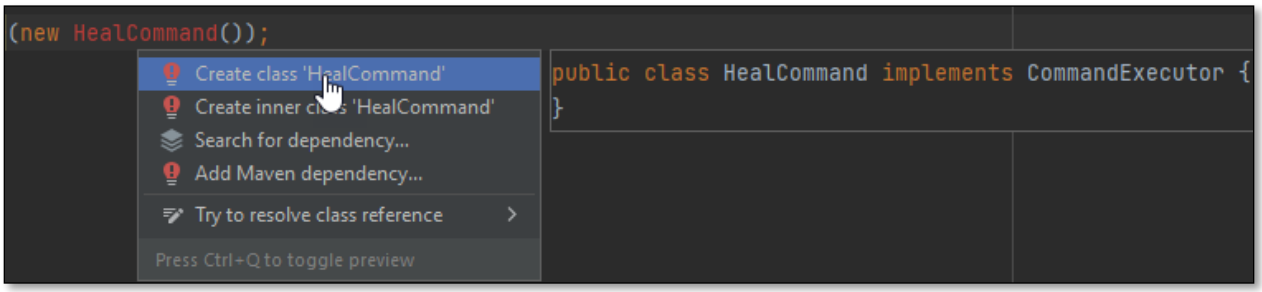
This adds the command **heal** to the plugin manifest. This makes the command available to the player in game.

10. Go back to **Heal.java**
11. Modify the exiting **onEnable()** method.

```java
@Override
public void onEnable() {
    // When the plugin is enabled
    // Create the heal command object
    getCommand("heal").setExecutor(new HealCommand());
}
```

12. HealCommand is red. Click **HealCommand()** → press **ALT ENTER**.

13. Choose Create class **HealCommand**. Click **OK**.



14. This will create a **HealCommand.java** file.

15. Again, more red marks. Click **CommandExecutor** → press **ALT ENTER**

16. Click **Implement Methods.** Click **OK**.

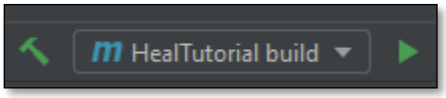
17. Add the following code to the existing **onCommand()** method.

```java
@Override
public boolean onCommand(CommandSender sender, Command comman
    // Is the command sender a Player?
    if(sender instanceof Player){
        // Create a player object
        Player player = (Player) sender;
        player.sendMessage("Your health has been restored!");
        // Set player health to 20 (100%);
        player.setHealth(20);
    }
    return false;
}
```

18. More red! Click **Player** → **ALT ENTER**. This should add the Player import. No more red.

5: Build and Test Plugin

Time to build our plugin jar file.



1. You should see the above toolbar in your project.
2. Click the hammer to Build Project. The build should be successful.
3. Click the Green go button to Run the project. This will create the jar file.
4. In File Explorer → **project folder** → **target**
5. There should be 2 jar files. Copy the one that starts with the name **heal-1.0-SNAPSHOT.jar**
6. Go to your Minecraft server folder → **plugins** → paste your jar file.
7. Start your Minecraft server. Watch for errors on startup. You should see your plugin enable.
8. Connect with your Minecraft client.
9. In Chat → **/heal**
10. You should be healed!
11. Change to Survival mode and incur some damage. Try your heal command again.

Congratulations, you have created your first Minecraft Server Plugin!

Assignment Submission

1. Insert screenshots showing your heal command in Minecraft.
2. Attach and submit in Blackboard.